UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2022

Roll No. ....

Paper: Statistical Computer Packages | **Course Code: STAT-309** | **Time: 3 Hrs.** | **Marks: 60**

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

NOTE: USE SPSS SOFTWARE TO SOLVE YOUR QUESTIONS (WRITE DETAILED PROCEDURE OF SOLUTION OF COMPUTER IS NOT AVAILABLE DURING EXAM).

Q.1. Answer the following short questions: (6x5=30)

i) Write benefits of defining value labels in SPSS data file.

ii) Explain the rules for giving names to variables in SPSS.

iii) If you have a variable x, then using SPSS how would to compute a new variable $y = \log(x)$?

iv) While entering data in SPSS, we have to give nominal/ordinal/scale in "measure" column for each variable, explain the different between nominal, ordinal and scale?

v) Write the procedure of applying a one-sample t-test in SPSS.

vi) Suppose you have a data on variables x_1, x_2, x_3 *and* x_4. Write procedure to compute partial correlation between x_1 *and* x_2.

Answers the following questions.

Q.2: Over a period of 40 days the percentage relative humidity in a vegetable storage building was measured. Mean daily values were recorded as shown below:

60	63	64	71	67	73	79	80	83	81
86	90	96	98	98	99	89	80	77	78
71	79	74	84	85	82	90	78	79	79
78	80	82	83	86	81	80	76	66	74

Find the median, lower quartile, upper quartile, arithmetic mean, variance, standard deviation and coefficient of variation from above sample of observations (08)

Q.3: CGPA of four classes of the students of B.S (Hons) 1st semester, session 2013-2017 programs were selected at random. Test the hypothesis that the mean CGPA of all four classes is equal. Using $\alpha = 0.05$. Interpret the P- value. (10)

Mathematics	Statistics	Chemistry	Physics
2.1	3.4	2.2	3.0
3.5	3.6	2.6	2.6
3.2	2.1	2.9	2.5
2.6	1.9	3.6	2.8
2.7	2.3	2.5	1.8

Q.4: It is generally believed that blood pressure is related to one's age and weight. To study the possible effects of age and weight on blood pressure, these data were recorded from a sample of 12 adult men:

Age (x_1)	48	50	35	41	33	25	51	53	40	31	35	46
Weight (x_2)	175	159	191	174	165	157	182	164	157	168	138	191
Blood Pressure (x_3)	143	131	135	131	121	115	133	126	120	128	120	160

Answer following questions (12)

a) Compute the regression equation for taking blood pressure as a response variable and weight & age as predictors.

b) Calculate standard error of estimate.

c) Compute and interpret the coefficient of determination.

d) Predict the blood pressure of adults who have age 40 and weight 190 lb.